

but there are no equidistant points above the graph of the function f . Therefore it is impossible to find different equidistant points along a vertical line. In particular, by choosing $y_1 = G(x) < y = y_2$, Lemma 1 shows that

$$\begin{aligned} d((x, y_2), L) - d((x, y_1), L) &< d((x, y_2), (x, y_1)) = y_2 - y_1 \Rightarrow \\ d((x, y), L) &< y = d((x, y), K) \end{aligned}$$

because equality of type (2) must be avoided. We can finish the proof by choosing $0 < y_1 = y < G(x) = y_2$ as follows:

$$\begin{aligned} d((x, y_2), L) - d((x, y_1), L) &< d((x, y_2), (x, y_1)) = y_2 - y_1 \Rightarrow \\ d((x, y), K) &= y < d((x, y), L) \end{aligned}$$

and the inequality is automatically satisfied in case of $y \leq 0$. $\square$

Definition 1. A function $G: \mathbb{R}^n \rightarrow \mathbb{R}^+$ is an equidistant function if its graph is the equidistant set of K and L for a positive-valued continuous function f .

Lemma 3. The equidistant function $G: \mathbb{R}^n \rightarrow \mathbb{R}^+$ belonging to a positive-valued continuous function $f: \mathbb{R}^n \rightarrow \mathbb{R}^+$ is continuous.

Proof. Taking $x_n \rightarrow x$, the sequence $y_n = G(x_n)$ is obviously bounded because of the continuity of the function f : for all but finitely many indices,

$$0 < y_n = G(x_n) < f(x_n) < f(x) + \varepsilon.$$

Therefore it has a convergent subsequence. Since the distance-measuring functions $d((x, y), L)$ and $d((x, y), K)$ are continuous, it can easily be seen that any convergent subsequence of y_n gives a subsequence of (x_n, y_n) tending to a point (x, y) such that

$$d((x, y), L) = d((x, y), K).$$

The unicity of the equidistant point at x implies that $y = G(x)$ is the common limit of the convergent subsequences of $y_n = G(x_n)$. Therefore $y_n = G(x_n) \rightarrow y = G(x)$. $\square$

Theorem 1. Let I be a finite nonempty index set and consider the family $\{f_i \mid i \in I\}$ of positive-valued continuous functions defined on $\mathbb{R}^n$ with corresponding equidistant functions $\{G_i \mid i \in I\}$. If $G_{\min}$ is the equidistant function belonging to the function

$$f_{\min}: \mathbb{R}^n \rightarrow \mathbb{R}, \quad f_{\min}(x) = \min \{f_i(x) \mid i \in I\},$$

then

$$G_{\min}(x) = \min \{G_i(x) \mid i \in I\}$$

for any $x \in \mathbb{R}^n$.

Proof. Since all functions are positive-valued, we can use the epigraphs as focal sets in the sense of Remark 1. Let $x \in \mathbb{R}^n$ be an arbitrary point, $L_i := \text{epi } f_i$ ($i \in I$) and suppose that

$$y > \min \{G_i(x) \mid i \in I\}.$$

Then there exists at least one index $i \in I$ such that $y > G_i(x)$ and Lemma 2 implies that $d((x, y), L_i) < d((x, y), K)$. Since $L := \text{epi } f_{\min} = \cup_{i \in I} \text{epi } f_i = \cup_{i \in I} L_i$, we have

$$d((x, y), L) = d((x, y), \cup_{i \in I} L_i) \leq d((x, y), L_i) < d((x, y), K)$$

and Lemma 2 implies that $y > G_{\min}(x)$. On the other hand, suppose that

$$y < \min \{G_i(x) \mid i \in I\}.$$

Then $y < G_i(x)$ for all $i \in I$ and Lemma 2 implies that $d((x, y), K) < d((x, y), L_i)$ for all $i \in I$. Thus

$$d((x, y), K) < \min \{d((x, y), L_i) \mid i \in I\} = d((x, y), \cup_{i \in I} L_i) = d((x, y), L)$$